



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**NATIONAL
SENIOR CERTIFICATE/
NASIONALE
SENIOR SERTIFIKAAT**

GRADE/GRAAD 12

**PHYSICAL SCIENCES: CHEMISTRY (P2)
FISIESE WETENSKAPPE: CHEMIE (V2)**

NOVEMBER 2023

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

**These marking guidelines consist of 19 pages.
Hierdie nasienriglyne bestaan uit 19 bladsye.**



QUESTION 1/VRAAG 1

- 1.1 D ✓✓ (2)
- 1.2 D ✓✓ (2)
- 1.3 A ✓✓ (2)
- 1.4 B ✓✓ (2)
- 1.5 D ✓✓ (2)
- 1.6 C ✓✓ (2)
- 1.7 C ✓✓ (2)
- 1.8 A ✓✓ (2)
- 1.9 B ✓✓ (2)
- 1.10 D ✓✓ (2)
- [20]**

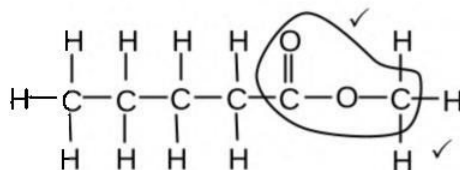
QUESTION 2/VRAAG 2

- 2.1 Molecules containing carbon atoms. ✓
Molekule wat koolstofatome bevat. (1)
- 2.2
- 2.2.1 2,3-dimethyl✓but-1-ene✓/2,3-dimethyl-1-butene
2,3-dimetielbut-1-een/2,3-dimetiel-1-buteen
- | | |
|--|---|
| Marking criteria: <ul style="list-style-type: none"> • Correct stem i.e. <u>but-1-ene</u>. ✓ • IUPAC name completely correct including numbering, sequence, hyphens and commas. ✓ | Nasienkriteria: <ul style="list-style-type: none"> • Korrekte stam d.i. <u>but-1-ene</u>. ✓ • IUPAC-naam heeltemal korrek insluitende nommering, volgorde, koppeltekens en kommas. ✓ |
|--|---|
- (2)
- 2.2.2 Butan-2-one/2-butanone/butanone ✓✓
Butan-2-oon/2-butanoon/butanoon (2)



2.3

2.3.1

**Marking criteria/Nasienkriteria:**

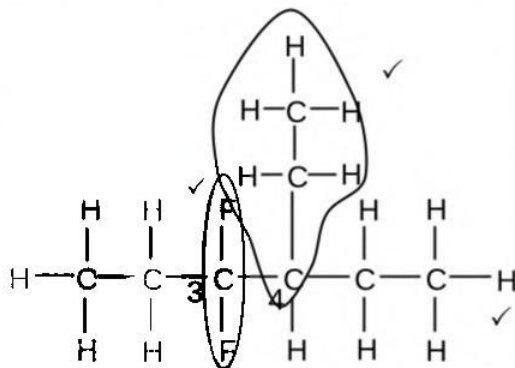
- Functional group correct ✓
Funksionele groep korrek.
- Whole structure correct. ✓
Hele struktuur korrek.

IF/INDIEN

- More than one functional group/wrong functional group:
Meer as een funksionele groep/foutiewe funksionele groep: $\frac{0}{2}$
- If condensed structural formulae used/*Indien gekondenseerde struktuurformules gebruik:* Max./Maks. $\frac{1}{2}$

(2)

2.3.2

**Marking criteria/Nasienkriteria:**

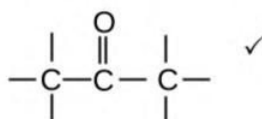
- Six C atoms in longest chain. ✓
Ses C-atome in langste ketting.
- Two F atoms on third C atom. ✓
Twee F-atome op die derde C-atoom.
- Ethyl substituent on fourth C atom. ✓
Etielsubstituent op die vierde C-atoom.

(3)

2.3.3 C_nH_{2n} ✓

(1)

2.3.4



(1)

2.3.5 Methanol/Metanol ✓✓

(2)

2.4.1 B ✓

(1)

2.4.2 D and/en G ✓

(1)

[16]

QUESTION 3/VRAAG 3

3.1

Marking criteria/Nasienkriteria

If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark. / Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

The underlined phrases must be in the correct context. / Die onderstreepte frases moet in die korrekte konteks wees.

The temperature at which the vapour pressure of a substance equals atmospheric pressure. ✓✓

Die temperatuur waarby die dampdruk van die stof gelyk is aan atmosferiese druk.

(2)

3.2

Marking criteria/Nasienkriteria:

- Dependent and independent variables correctly identified. ✓
Afhanklike en onafhanklike veranderlikes korrek geïdentifiseer.
- Correct relationship between dependent and independent variables stated. ✓
Korrekte verwantskap tussen die afhanklike en onafhanklike veranderlikes gestel

The higher the molecular mass the higher the boiling point. / As the molecular mass increases the boiling point increases. / The longer the C-chain length the higher boiling point / The boiling point and the molecular mass are proportional. ✓✓

Hoe hoër die molekulêre massa hoe hoër die kookpunt. / Soos die molekulêre massa toeneem, neem die kookpunt ook toe. / Hoe langer die C-kettinglengte hoe hoër is die kookpunt. / Die kookpunt en die molekulêre massa is eweredig.

(2)

3.3

Marking criteria:

- Compare the strength of intermolecular forces. ✓
- Compare the energy required to overcome intermolecular forces. ✓

Nasienkriteria:

- Vergelyk die sterkte van intermolekulêre kragte. ✓
- Vergelyk die energie benodig om intermolekulêre kragte te oorkom. ✓

• Strength of the intermolecular forces increases / More sites for London forces with increase of molar mass/chain length. ✓

• More energy is needed to overcome/break intermolecular forces. ✓

OR

• Strength of the intermolecular forces decreases / Less sites for London forces with decrease in molar mass/chain length. ✓

• Less energy is needed to overcome/break intermolecular forces. ✓

• Sterkte van die intermolekulêre kragte verhoog / Meer punte vir Londonkragte met 'n toename in molêre massa/kettinglengte. ✓

• Meer energie benodig om intermolekulêre kragte te oorkom/breek. ✓

OF

• Sterkte van die intermolekulêre kragte verlaag / Minder punte vir Londonkragte met 'n afname in molêre massa/kettinglengte. ✓

• Minder energie benodig om intermolekulêre kragte te oorkom/breek. ✓

(2)



3.4.1 Aldehyde / Aldehiede ✓

(1)

3.4.2 **Marking criteria:**

- Comparing the strength of intermolecular forces of aldehydes with alcohols and/or carboxylic acids. ✓
- Linking the intermolecular forces to boiling point. ✓

Nasienkriteria:

- Vergelyk die sterkte van die intermolekulêre kragte van aldehiede met alkohole en/of karboksielsure. ✓
- Trek die verband tussen die intermolekulêre kragte en die kookpunte. ✓
- The strength of the intermolecular forces in aldehydes is weaker than in alcohols / carboxylic acids. ✓
- Therefore aldehydes have lower boiling points than alcohols / carboxylic acids ✓

OR

- Carboxylic acids and alcohols have stronger intermolecular forces than aldehydes.
- Therefore carboxylic acids and/or alcohols have higher boiling points than aldehydes.
- Die sterkte van die intermolekulêre kragte tussen aldehiede is swakker as tussen alkohole / karboksielsure. ✓
- Dus het aldehiede 'n laer kookpunt as alkohole / karboksielsure. ✓

OF

- Karboksielsure en alkohole het sterker intermolekulêre kragte as aldehiede
- Dus het alkohole / karboksielsure 'n hoër kookpunt as aldehiede.

(2)

3.5

3.5.1 60 (g·mol⁻¹) ✓

(1)

3.5.2 **POSITIVE MARKING FROM QUESTION 3.4/POSITIEWE NASIEN VAN VRAAG 3.4**

Propan-1-ol/1-propanol ✓✓

Marking criteria:

- Correct chain length, 3 C-atoms ✓
- Correct IUPAC name ✓

Nasienkriteria:

- Korrekte stamlengte, 3 C-atome. ✓
- Korrekte IUPAC-naam. ✓

(2)



3.6

Marking criteria:

- State that carboxylic acids have two sites for hydrogen bonding. ✓
- State that alcohols have one site for hydrogen bonding. ✓
- Comparing the strength of IMF's / the energy needed to overcome IMF's. ✓

Nasienkriteria:

- Stel dat karboksielsure twee plekke het vir waterstofbindings.
- Stel dat alkohole een plek het vir waterstofbinding.
- Vergelyk die sterkte van die IMK's / energie benodig om IMK's te oorkom.

- Carboxylic acids/B have, in addition to London forces and dipole-dipole forces, two sites for hydrogen bonding between molecules. ✓

OR

Carboxylic acids can form dimers due to strong hydrogen bonding between molecules.

- Alcohols/A have, in addition to London forces and dipole-dipole forces, one site for hydrogen bonding between molecules. ✓
- Intermolecular forces in carboxylic acids are stronger./More energy needed to overcome/break intermolecular forces in carboxylic acid/B. ✓
- *Karboksielsure het, in toevoeging tot Londonkragte en dipool-dipoolkragte, twee punte vir waterstofbinding tussen molekule.*

OF

Karboksielsure kan dimere vorm as gevolg van sterk waterstofbindings tussen molekule.

- *Alkohole het, in toevoeging tot Londonkragte en dipool-dipoolkragte, een punt vir waterstofbinding tussen molekule.*
- *Intermolekulêre kragte in karboksielsure is sterker./Meer energie word benodig om intermolekulêre kragte in karboksielsure te oorkom/breek.*

(3)
[15]

QUESTION 4/VRAAG 4

4.1

4.1.1

Marking criteria/Nasienkriteria

If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark. / Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

The underlined phrases must be in the correct context. / Die onderstreepte frases moet in die korrekte konteks wees.

The chemical process in which longer chain hydrocarbon molecules are broken down to shorter (more useful) molecules. ✓✓

Die chemiese proses waarin langer kettingkoolwaterstof-molekule afgebreek word in korter, (meer bruikbare), molekules.

(2)

4.1.2

X = 12 ✓

Y = 2 ✓

Z = 4 ✓

(3)

4.1.3

Marking criteria/Nasienkriteria

- Reactants ✓ products ✓ / Reaktanse produkte
- Balancing ✓ / Balansering

**Notes/Aantekeninge:**

- Ignore double arrows and phases. / Ignoreer dubbelpyle en fases.
- Marking rule 6.3.10 / Nasienreël 6.3.10.
- If condensed structural formulae used: / Indien gekondenseerde struktuurformules gebruik: Max/Maks. $\frac{2}{3}$

(3)

4.2

4.2.1

Marking criteria/Nasienkriteria

If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark. / Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

The underlined phrases must be in the correct context. / Die onderstreepte frases moet in die korrekte konteks wees.

Compounds with the same molecular formula, but different positions of the side chain, substituents or functional groups on the parent chain. ✓✓

Verbindings met dieselfde molekulêre formule, maar verskillende posisies van die syketting, substituenten of funksionele groepe op die stamketting.

(2)

4.2.2

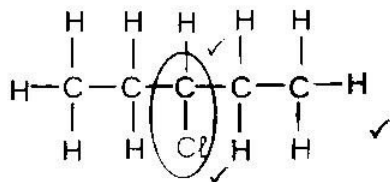
Addition/hydrohalogenation/hydrochlorination ✓

Addisie/hidrohalogenering/hidrochlorinering

(1)



4.2.3

**Marking criteria/Nasienkriteria:**

- Chlorine atom bonded to any C-atom ✓
Chlooratoom gebind aan enige C-atoom
- Correct functional group on third C-atom ✓
Korrekte funksionele groep op derde C-atoom
- Whole structure correct ✓
Hele struktuur korrek

(3)

4.2.4 HCl ✓

(1)

4.2.5 Concentrated sulphuric acid/H₂SO₄(conc./gek.) ✓Gekonsentreerde swawelsuur

(1)

4.2.6 Concentrated strong base ✓**OR**Concentrated NaOH/KOH/LiOH/sodium hydroxide/ potassium hydroxide/ lithium hydroxide**OR**Strong base/NaOH/KOH/LiOH/sodium hydroxide/ potassium hydroxide/lithium hydroxide in ethanol.Gekonsentreerde sterk basis**OF**Gekonsentreerde NaOH /KOH/ LiOH /natriumhidroksied/ kaliumhidroksied/ litiumhidroksied**OF**Sterk basis/NaOH /KOH/ LiOH / natriumhidroksied/kaliumhidroksied/litiumhidroksied in etanol

(1)

4.2.7 Elimination ✓

Dehydrohalogenation/dehydrochlorination ✓

*Eliminasie**Dehidrohalogenering/dehidrohalogenasie/dehidrochlornasie/ dehidrochlonerig*

(2)

[19]

QUESTION 5/VRAAG 5

5.1

ANY ONE:

- Change in concentration ✓ of products/reactants per (unit) time. ✓
- Change in amount/number of moles/volume/mass of products or reactants per (unit) time.
- Amount/number of moles/volume/mass of products formed/reactants used per (unit) time.
- Rate of change in concentration/amount/number of moles/volume/mass. ✓✓ **(2 or 0)**

ENIGE EEN:

- Verandering in konsentrasie van produkte/reaktanse per (eenheid) tyd.
- Verandering in hoeveelheid/getal mol/volume/massa van produkte of reaktanse per (eenheid) tyd.
- Hoeveelheid/getal mol/volume/massa van produkte gevorm/reaktanse gebruik per (eenheid) tyd.
- Tempo van verandering in konsentrasie/ hoeveelheid/getal mol/volume/massa. **(2 of 0)**

(2)

5.2

Concentration (of $\text{Na}_2\text{S}_2\text{O}_3$)/Konsentrasie van ($\text{Na}_2\text{S}_2\text{O}_3$) ✓

(1)



5.3

Marking criteria/Nasienkriteria:

- Substitute/Vervang $0,03 \times 0,13 / 30 \times 0,13$ ✓
- Substitute/Vervang 50 OR/OF 0,05 ✓
- Final correct answer/Finale korrekte antwoord: $0,078 \text{ mol} \cdot \text{dm}^{-3}$ ✓
Range 0,075 to $0,08 \text{ mol} \cdot \text{dm}^{-3}$

OPTION 1/OPSIE 1

$$c = \frac{n}{V}$$

$$0,13 = \frac{n}{0,03} \quad \checkmark$$

$$n = 3,9 \times 10^{-3} \text{ moles/mol}$$

$$c = \frac{n}{V}$$

$$c = \frac{3,9 \times 10^{-3}}{0,05} \quad \checkmark$$

$$= 0,078 \text{ mol} \cdot \text{dm}^{-3} \quad \checkmark$$

OPTION 2/OPSIE 2

$$c_1 V_1 = c_2 V_2$$

$$(0,13)(0,030) \quad \checkmark = c_2 (0,50) \quad \checkmark$$

$$c_2 = 0,078 \text{ mol} \cdot \text{dm}^{-3} \quad \checkmark$$

OPTION 3/OPSIE 3**Marking criteria/Nasienkriteria:**

- Substitute/Vervang $0,05 \times 0,13$ OR/OF $50 \times 0,13$ ✓
- Substitute/Vervang 50 OR/OF 0,05 ✓
- Final correct answer/Finale korrekte antwoord: $0,078 \text{ mol} \cdot \text{dm}^{-3}$ ✓
Range 0,075 to $0,08 \text{ mol} \cdot \text{dm}^{-3}$

$$c = \frac{n}{V}$$

$$0,13 = \frac{n}{0,05} \quad \checkmark$$

$$n = 6,5 \times 10^{-3} \text{ moles/mol}$$

$$V_2 : V_1$$

$$3 : 5$$

$$3,9 \times 10^{-3} : 6,5 \times 10^{-3}$$

$$c = \frac{n}{V}$$

$$c = \frac{3,9 \times 10^{-3}}{0,05} \quad \checkmark$$

$$= 0,078 \text{ mol} \cdot \text{dm}^{-3} \quad \checkmark$$

OR/OF

$$c = \frac{n}{V}$$

$$0,10 = \frac{n}{0,05} \quad \checkmark$$

$$n = 5 \times 10^{-3} \text{ moles/mol}$$

$$V_2 : V_1$$

$$3 : 4$$

$$3,75 \times 10^{-3} : 5 \times 10^{-3}$$

$$c = \frac{n}{V}$$

$$c = \frac{3,75 \times 10^{-3}}{0,05} \quad \checkmark$$

$$= 0,075 \text{ mol} \cdot \text{dm}^{-3} \quad \checkmark$$

OPTION 4/OPSIE 4

$$\frac{3}{5} \quad \checkmark \times 0,13 \quad \checkmark = 0,078 \text{ mol} \cdot \text{dm}^{-3} \quad \checkmark$$

(3)



5.4

Marking criteria:

- Substitute to calculate $n(S)$: $\frac{0,21}{32}$ ✓
- Use mol/M ratio: $n(S) = n(Na_2S_2O_3)$ ✓
- Substitute $M = 158 \text{ g} \cdot \text{mol}^{-1}$ in formula
 $n(Na_2S_2O_3) = \frac{m}{M}$ ✓
- Substitute $t = 20,4 \text{ s}$ into rate formula.
 ✓
- Final correct answer: $0,051 \text{ (g} \cdot \text{s}^{-1})$ ✓
 Range: $0,050$ to $0,080 \text{ (g} \cdot \text{s}^{-1})$

Nasiënkrriteria:

- Vervang om te bereken $n(S)$: $\frac{0,21}{32}$ ✓
- Gebruik mol/M-verhouding:
 $n(S) = n(Na_2S_2O_3)$ ✓
- Vervang $M = 158 \text{ g} \cdot \text{mol}^{-1}$ in formula
 $n(Na_2S_2O_3) = \frac{m}{M}$ ✓
- Vervang $t = 20,4 \text{ s}$ in tempo formule. ✓
- Finale korrekte antwoord: $0,051 \text{ (g} \cdot \text{s}^{-1})$ ✓
 Range/Gebied: $0,050$ to $0,080 \text{ (g} \cdot \text{s}^{-1})$

OPTION 1/OPSIE 1

$$n(S) = \frac{m}{M}$$

$$= \frac{0,21}{32} \quad \checkmark$$

$$= 0,00656 \text{ moles/mol (} 6,56 \times 10^{-3} \text{)}$$

$$n(S) = n(Na_2S_2O_3)$$

$$= 0,00656 \text{ moles/mol } \checkmark$$

$$n(Na_2S_2O_3) = \frac{m}{M}$$

$$0,00656 = \frac{m}{158} \quad \checkmark$$

$$m(Na_2S_2O_3) = 1,04 \text{ g}$$

OPTION 2/OPSIE 2

$$158 \text{ g } Na_2S_2O_3 \checkmark \longrightarrow 32 \text{ g S } \checkmark$$

$$x \text{ g} \longrightarrow 0,21 \text{ g } \checkmark$$

$$x = 1,04 \text{ g}$$

$$\text{Rate/Tempo} = \frac{\Delta m}{\Delta t}$$

$$= \frac{1,04}{20,4} \quad \checkmark$$

$$= 0,051 \text{ (g} \cdot \text{s}^{-1}) \quad \checkmark$$

ACCEPT/AANVAAR:

$$c = \frac{n}{V}$$

$$0,13 = \frac{n}{0,05}$$

$$= 0,00656$$

$$n(Na_2S_2O_3) = \frac{m}{M}$$

$$0,00656 = \frac{m}{158} \quad \checkmark$$

$$= 1,03 \text{ g (} 1,027 \text{)}$$

$$\text{Rate/Tempo} = \frac{\Delta m}{\Delta t}$$

$$= \frac{1,03}{20,4} \quad \checkmark$$

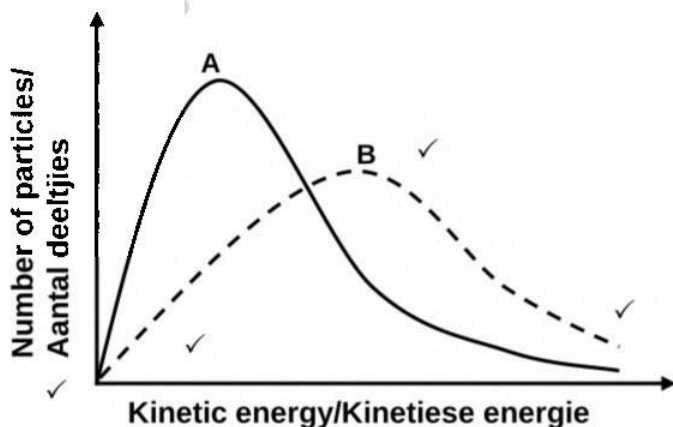
$$= 0,05 \text{ (g} \cdot \text{s}^{-1}) \quad \checkmark$$

Max/Maks. $\frac{3}{5}$

(5)



5.5

**Marking criteria/Nasienkriteria:**

- Both axis labelled correctly. ✓
Beide asse korrek benoem
- Both curves start at origin and have correct shape. ✓
Beide kurwes begin by die oorsprong en het dieselfde vorm.
- Peak of curve B must be lower than curve A. ✓
Maksimum van kurwe B moet laer wees as kurwe A.
- Peak of curve B must have higher kinetic energy than curve A up to end of curve. ✓
Maksimum van kurwe B moet hoër wees as kinetiese energie van kurwe A tot by die einde.

(4)

5.6

- At a higher temperature particles move faster/have higher kinetic energy. ✓
- More molecules have enough/sufficient kinetic energy for an effective collision. ✓
OR More molecules have kinetic energy/ E_k equal to or greater than the activation energy.
- More effective collisions per unit time/second. ✓
OR Frequency of effective collisions increases.
- Reaction rate increases. ✓
- *By 'n hoër temperatuur beweeg die deeltjies vinniger/het die deeltjies hoër kinetiese energie.* ✓
- Meer molekule het genoeg/voldoende kinetiese energie/ E_k vir 'n effektiewe botsing. ✓
OF Meer molekule het kinetiese energie gelyk aan of groter as die aktiveringsenergie.
- Meer effektiewe botsings per eenheidtyd/sekonde. ✓
OF Frekwensie van effektiewe botsings verhoog.
- Reaksietempo neem toe. ✓

(4)
[19]

QUESTION 6/VRAAG 6

- 6.1 A reaction is reversible when products can be converted back to reactants (and vice versa). ✓

'n Reaksie is omkeerbaar wanneer produkte terug na reaktanse, en (omgekeerd), omgeskakel kan word.

(1)

- 6.2 **Marking criteria/Nasienkriteria:**

If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark./Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

The underlined phrases must be in the correct context. / Die onderstreepte frases moet in die korrekte konteks wees.

When the equilibrium in a closed system is disturbed, the system will re-instate a new equilibrium by favouring the reaction that will cancel/oppose the disturbance. ✓✓

Wanneer die ewewig in 'n geslote sisteem versteur word, sal die sisteem 'n nuwe ewewig instel deur die reaksie te bevoordeel wat die versteuring kanselleer/teenwerk.

(2)

- 6.3.1 The amount of $A_2(g)$ was increased./ A_2 was added into the container. ✓

Die hoeveelheid $A_2(g)$ is verhoog./ A_2 is bygevoeg in die houer.

(1)

- 6.3.2
- Increase in A_2 favours the reaction that uses or decreases the amount/concentration of A_2 . ✓
 - The reverse reaction is favoured/amount or concentration of products decreases/amount or concentration of reactants increases. ✓

- 'n Toename in A_2 bevoordeel die reaksie wat die hoeveelheid/konsentrasie van A_2 verlaag
- Die terugwaartse reaksie is bevoordeel/hoeveelheid of konsentrasie van die produkte neem af/die hoeveelheid of konsentrasie van die reaktante neem toe.

(2)

- 6.4 **Marking criteria/Nasienkriteria:**

- Correct K_c expression./ Korrekte K_c -uitdrukking. ✓
- Substitution of $[A_2]$ and $[B_2]$ at equilibrium. ✓
Substitusie van $[A_2]$ en $[B_2]$ by ewewig.
- Substitution of $[AB]$ at equilibrium. ✓
Substitusie van $[AB]$ by ewewig.
- Final correct answer/Finale korrekte antwoord: 0,16✓

IF/INDIEN:

Wrong or no K_c expression:

Verkeerde of geen K_c - uitdrukking: Max./Maks. $\frac{2}{4}$

Moles substituted/Mol vervang Max./Maks. $\frac{2}{4}$

OPTION 1/OPSIE 1

$$K_c = \frac{[A_2][B_2]}{[AB]^2} \checkmark$$

$$= \frac{\left(\frac{8}{4}\right)\left(\frac{2}{4}\right)}{\left(\frac{10}{4}\right)^2} \checkmark$$

$$= 0,16 \checkmark$$

OPTION 2/OPSIE 2

$$K_c = \frac{[A_2][B_2]}{[AB]^2} \checkmark$$

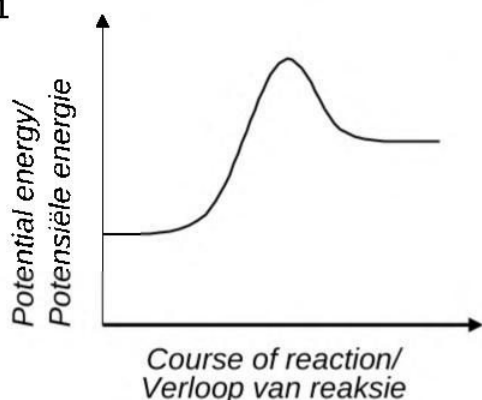
$$= \frac{(2)(0,5)}{(2,5)^2} \checkmark$$

$$= 0,16 \checkmark$$

(4)



6.5.1

**Marking criteria/Nasienkriteria:**

- Both axes correctly labelled and correct shape of Ep curve. ✓
Asse korrek benoem en korrekte vorm van Ep-kurwe
- Shape of Ep curve for endothermic reaction as shown. ✓✓
Vorm van kurwe vir endotermiese reaksie soos getoon.

(3)

6.5.2 • Smaller than ✓

- Products/ B_2/A_2 amount/concentration decreases. ✓
- Reactants/AB amount/concentration increases. ✓

OR

The reverse reaction is favoured. / Equilibrium (position) shifts to the left. ✓✓

- Kleiner as*

- Produkte/ B_2/A_2 hoeveelheid/ konsentrasie neem af.*
- Reaktanse/AB hoeveelheid/konsentrasie neem toe.*

OF*Die terugwaartse reaksie word bevoordeel./Die ewewigs(posisie) skuif na links.*

(3)

6.6

- Both forward and reverse reaction rates increase equally. / Gradient of three curves will be steeper. ✓✓
- Reaches equilibrium sooner/less than 40 s. / The graph becomes horizontal sooner. ✓
- Beide die voorwaartse en terugwaartse reaksietempo verhoog dieselfde. / Gradiënt van al drie kurwes is steiler.*
- Ewewig word vinniger/korter tyd/minder as 40 s bereik. / Grafiek neem korter tyd om horisontaal te word.*

(3)

[19]

QUESTION 7/VRAAG 7

- 7.1 A strong base ionises/dissociates completely ✓ in water. ✓
 Sterk basis ioniseer/dissosieer volledig in water.

(2)

7.2.1 $n(\text{Ba}(\text{OH})_2) = cV$ ✓
 $= \frac{0,15}{1000} \times 0,02$ ✓
 $= 0,003 \text{ mol}$ ✓

(3)

7.2.2 **POSITIVE MARKING FROM QUESTION 7.2.1/**
POSITIEWE NASIEN VAN VRAAG 7.2.1

Marking criteria:	Nasiën kriteria:
(a) Use ratio: $2n\text{Ba}(\text{OH})_2$ (7.2.1) = $n\text{HNO}_3$ ✓ (b) Substitute $n\text{H}_3\text{O}^+$ or $n\text{HNO}_3$ and $0,045 \text{ dm}^3$ in $c = \frac{n}{V}$ ✓ (c) Formula: $\text{pH} = -\log[\text{H}_3\text{O}^+]$ ✓ (d) Substitute $[\text{H}_3\text{O}^+]$ in pH formula ✓ (e) Final correct answer: 0,89 ✓ Range: 0,88 to 0,89	(a) Gebruik verhouding: $2n\text{Ba}(\text{OH})_2$ (7.2.1) = $n\text{HNO}_3$ ✓ (b) Vervang $n\text{H}_3\text{O}^+$ of $n\text{HNO}_3$ en $0,045 \text{ dm}^3$ in $c = \frac{n}{V}$ ✓ (c) Formule: $\text{pH} = -\log[\text{H}_3\text{O}^+]$ ✓ (d) Vervang $[\text{H}_3\text{O}^+]$ in pH formule ✓ (e) Finale korrekte antwoord: 0,89 ✓ Gebied: 0,88 tot 0,89
$n\text{HNO}_3 \text{ reacted} = 2n\text{Ba}(\text{OH})_2$ $= 2(0,003)$ $= 0,006 \text{ mol}$ ✓(a)	
OPTION 1/ OPSIE 1 $n(\text{H}_3\text{O}^+) = n(\text{HNO}_3)$ $= 0,006 \text{ mol}$ $[\text{H}_3\text{O}^+] = \frac{n}{V}$ $= \frac{0,006}{0,045}$ ✓(b) $= 0,13 \text{ mol} \cdot \text{dm}^{-3}$	OPTION 2/ OPSIE 2 $[\text{HNO}_3] = \frac{n}{V}$ $= \frac{0,006}{0,045}$ ✓(b) $= 0,13 \text{ mol} \cdot \text{dm}^{-3}$ $[\text{H}_3\text{O}^+] = [\text{HNO}_3]$ $= 0,13 \text{ mol} \cdot \text{dm}^{-3}$
$\text{pH} = -\log[\text{H}_3\text{O}^+]$ ✓(c) $= -\log(0,13)$ ✓(d) $= 0,89$ ✓(e)	

(5)



7.3

POSITIVE MARKING FROM QUESTION 7.2.2/**POSITIEWE NASIEN VAN VRAAG 7.2.2**

Marking criteria:	Nasienkriteria:
(a) Substitute $[\text{HNO}_3] = 0,4 \text{ mol}\cdot\text{dm}^{-3}$ and $0,025 \text{ dm}^3$ ✓	(a) Vervang: $[\text{HNO}_3] = 0,4 \text{ mol}\cdot\text{dm}^{-3}$ en $0,025 \text{ dm}^3$ ✓
(b) Subtract: $n(\text{HNO}_3)_{\text{ini}} - n(\text{HNO}_3)_{\text{excess}}$ (7.2.2) ✓ ✓	(b) Trek af: $n(\text{HNO}_3)_{\text{aanv}} - n(\text{HNO}_3)_{\text{oormaat}}$ (7.2.2) ✓ ✓
(c) Use of ratio $n(\text{MCO}_3) = \frac{1}{2}n(\text{HNO}_3)$ ✓	(c) Gebruik verhouding: $n(\text{MCO}_3) = \frac{1}{2}n(\text{HNO}_3)$ ✓
(d) Calculate the pure $m(\text{MCO}_3)$ ✓	(d) Bereken suiwer $m(\text{MCO}_3)$ ✓
(e) Substitute $n(\text{MCO}_3)$ and $m(\text{MCO}_3)$ in $n = \frac{m}{M}$ ✓	(e) Vervang $n(\text{MCO}_3)$ en $m(\text{MCO}_3)$ in $n = \frac{m}{M}$ ✓
(f) Calculation of $24 \text{ g}\cdot\text{mol}^{-1}$ ✓	(f) Berekening van $24 \text{ g}\cdot\text{mol}^{-1}$ ✓
(g) Correct answer: Mg ✓	(g) Korrekte antwoord: Mg ✓

$$\begin{aligned}
 n(\text{HNO}_3)_{\text{ini}} &= cV \\
 &= 0,4 \times 0,025 \text{ ✓ (a)} \\
 &= 0,01 \text{ mol}
 \end{aligned}$$

$$\begin{aligned}
 n(\text{HNO}_3)_{\text{react}} &= n(\text{HNO}_3)_{\text{ini}} - n(\text{HNO}_3)_{\text{excess}} \\
 &= 0,01 - 0,006 \text{ ✓ ✓ (b)} \\
 &= 0,004 \text{ mol}
 \end{aligned}$$

$$\begin{aligned}
 n(\text{MCO}_3) &= \frac{1}{2}n(\text{HNO}_3) \\
 &= \frac{1}{2}(0,004) \text{ ✓ (c)} \\
 &= 0,002 \text{ mol}
 \end{aligned}$$

$$\begin{aligned}
 m(\text{MCO}_3) &= \frac{85}{100} \times 0,198 \text{ ✓ (d)} \\
 &= 0,168 \text{ g}
 \end{aligned}$$

$$\begin{aligned}
 n(\text{MCO}_3) &= \frac{m}{M} \\
 0,002 &= \frac{0,168}{M} \text{ ✓ (e)}
 \end{aligned}$$

$$M(\text{MCO}_3) = 84 \text{ g}\cdot\text{mol}^{-1}$$

$$\begin{aligned}
 \text{Molar mass (M)} &= 84 - 60 \text{ ✓ (f)} \\
 &= 24 \text{ g}\cdot\text{mol}^{-1}
 \end{aligned}$$

Therefore metal M is Mg ✓ (g)

(8)
[18]



QUESTION 8/VRAAG 8

- 8.1.1 Copper strip becomes thinner/decreases in mass/solid/silver coloured particles in solution/the copper becomes plated with silver ✓

Koper plaatjie word dunner/massa neem af/vaste stof/silwer-kleurige deeltjies in oplossing.

(1)

- 8.1.2 Ag^+ ion/-ioon / Silver ion/Silwer-ioon ✓

(1)

- 8.2 Ag^+ ion is a stronger oxidising agent ✓ than Cu^{2+} ion ✓ and will oxidise Cu to (blue) Cu^{2+} ion. ✓

OR

Cu^{2+} ion is a weaker oxidising agent ✓ than Ag^+ ion ✓ and Cu will be oxidised to Cu^{2+} ion. ✓

OR

Cu/Copper is a stronger reducing agent ✓ than Ag/Silver ✓ and will reduce silver ions to silver. ✓

Ag^+ -ioon is 'n sterker oksideermiddel as Cu^{2+} -ioon en sal Cu na (blou) Cu^{2+} -ioon oksideer.

OF

Cu^{2+} -ioon is 'n swakker oksideermiddel as Ag^+ -ioon en daarom sal Cu na (blou) Cu^{2+} -ioon geoksideer word.

OF

Cu/Koper is 'n sterker reduseermiddel as Ag/Silwer en sal silwer-ione na silwer reduseer.

(3)

8.3

- 8.3.1 Silver/Ag/Silwer ✓

(1)

- 8.3.2 $\text{CuSO}_4/\text{Cu}^{2+}$ /Copper (II) ions/copper(II) sulphate/Koper(II)-ione/
koper(II)sulfaat ✓

(1)

ACCEPT/AANVAAR:

Any soluble copper(II) salt/Enige oplosbare koper(II)sout

- 8.3.3 $2\text{Ag}^+(\text{aq}) + \text{Cu}(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + \text{Cu}^{2+}(\text{aq})$ ✓ Bal ✓

Marking criteria/Nasienkriteria:

- Reactants ✓ Products ✓ Balancing: ✓
Reaktanse Produkte Balansering
- Ignore double arrows./Ignoreer dubbelpyle.
- Ignore phases./Ignoreer fases.
- Marking rule 6.3.10./Nasienreël 6.3.10.

(3)



8.4 K^+ ✓

The positive ions move to the silver ion solution to maintain the ion balance/electrical neutrality/ $[Ag^+]$ decreases. ✓

OR

It is the cathode.

OR

Positive charges decrease.

Die positiewe ione beweeg na die silwerioon-oplossing om ion-balans/elektriese neutraliteit te handhaaf/ $[Ag^+]$ neem af.

OF

Dit is die katode.

OF

Positiewe ladings verminder.

(2)
[12]**QUESTION 9/VRAAG 9**9.1 **ANY ONE/ENIGE EEN:**

- The chemical process in which electrical energy is converted to chemical energy. ✓✓

Die chemiese proses waarin elektriese energie omgeskakel word na chemiese energie.

- The use of electrical energy to produce a chemical change.

Die gebruik van elektriese energie om 'n chemiese verandering te weeg te bring.

- Decomposition of an ionic compound by means of electrical energy.

Ontbinding van 'n ioniese verbinding met behulp van elektriese energie.

- The process during which an electric current passes through a solution/ionic liquid/molten ionic compound.

Die proses waardeur 'n elektriese stroom deur 'n oplossing/ioniese vloeistof/gesmelte ioniese verbinding beweeg.

(2)

9.2 $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ ✓✓

Ignore phases/Ignoreer fases

Marking criteria/Nasienkriteria:

- $Cu(s) \leftarrow Cu^{2+}(aq) + 2e^-$ ($\frac{2}{2}$) $Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ ($\frac{1}{2}$)
 $Cu^{2+}(aq) + 2e^- \leftarrow Cu(s)$ ($\frac{0}{2}$) $Cu(s) \rightleftharpoons Cu^{2+}(aq) + 2e^-$ ($\frac{0}{2}$)
- Ignore if charge omitted on electron./Ignoreer indien lading weggelaat op elektron.
- If charge (+) omitted on Cu^{2+} /Indien lading (+) weggelaat op Cu^{2+} :
 Example/Voorbeeld: $Cu^2(aq) + 2e^- \rightarrow Cu(s)$ Max./Maks: $\frac{1}{2}$

(2)

9.3 R to/na Q ✓

(1)



9.4

Marking criteria:

- (a) Substitution of 63,5 into $n = \frac{m}{M}$ ✓
 (b) $N(\text{electrons}) = N(\text{Cu atoms}) \times 2$ ✓
 (c) Substitute $1,6 \times 10^{-19} \text{ C}$ in $n = \frac{Q}{e}$ ✓
 (d) Substitute (5)(60)(60) in $I = \frac{Q}{\Delta t}$ ✓
 (e) Final correct answer: 2,68 A ✓
 Range: 2,68 to 2,70 A

Nasienkriteria:

- (a) Vervang 63,5 in $n = \frac{m}{M}$ ✓
 (b) $N(\text{elektrone}) = N(\text{Cu-atome}) \times 2$ ✓
 (c) Vervang $1,6 \times 10^{-19} \text{ C}$ in $n = \frac{Q}{e}$ ✓
 (d) Vervang (5)(60)(60) in $I = \frac{Q}{\Delta t}$ ✓
 (e) Finale korrekte antwoord: 2,68 A ✓
 Gebied: 2,68 tot 2,70 A

$$\begin{aligned}
 n(\text{Cu}) &= \frac{m}{M} \\
 n(\text{Cu}) &= \frac{16}{63,5} \checkmark (\text{a}) \\
 &= 0,25 \text{ mol} \\
 n \text{ atoms}(\text{Cu}) &= \frac{N}{N_A} \\
 0,25 &= \frac{N}{6,02 \times 10^{23}} \\
 &= 1,5 \times 10^{23} \text{ atoms} \\
 N \text{ electrons} &= (1,5 \times 10^{23})(2) \checkmark (\text{b}) \\
 &= 3 \times 10^{23} \text{ electrons} \\
 N \text{ electrons} &= \frac{Q}{e} \text{ OR/OF } \frac{Q}{q_e} \\
 3 \times 10^{23} &= \frac{Q}{1,6 \times 10^{-19}} \checkmark (\text{c}) \\
 &= 48\,160 \text{ C} \\
 I &= \frac{Q}{\Delta t} \\
 &= \frac{48\,160}{(5)(60)(60)} \checkmark (\text{d}) \\
 &= 2,68 \text{ A} \checkmark (\text{e})
 \end{aligned}$$

(5)

9.5

Ag/silver is a weaker reducing agent ✓ than Cu/coper or Zn/zinc ✓ and will not be oxidised.

Ag/silwer is 'n swakker reduseermiddel as Cu/koper of Zn/sink en sal nie geoksideer word nie.

Voltage of power source is not effective enough to oxidise Ag/silver./ ✓✓

Die potensiaalverskil van die energiebron is nie effektief genoeg om die Ag/silwer te oksideer nie.

(2)

[12]

TOTAL/TOTAAL:

150

